MeV-GeV Counterparts to Fast Radio Bursts with VERITAS

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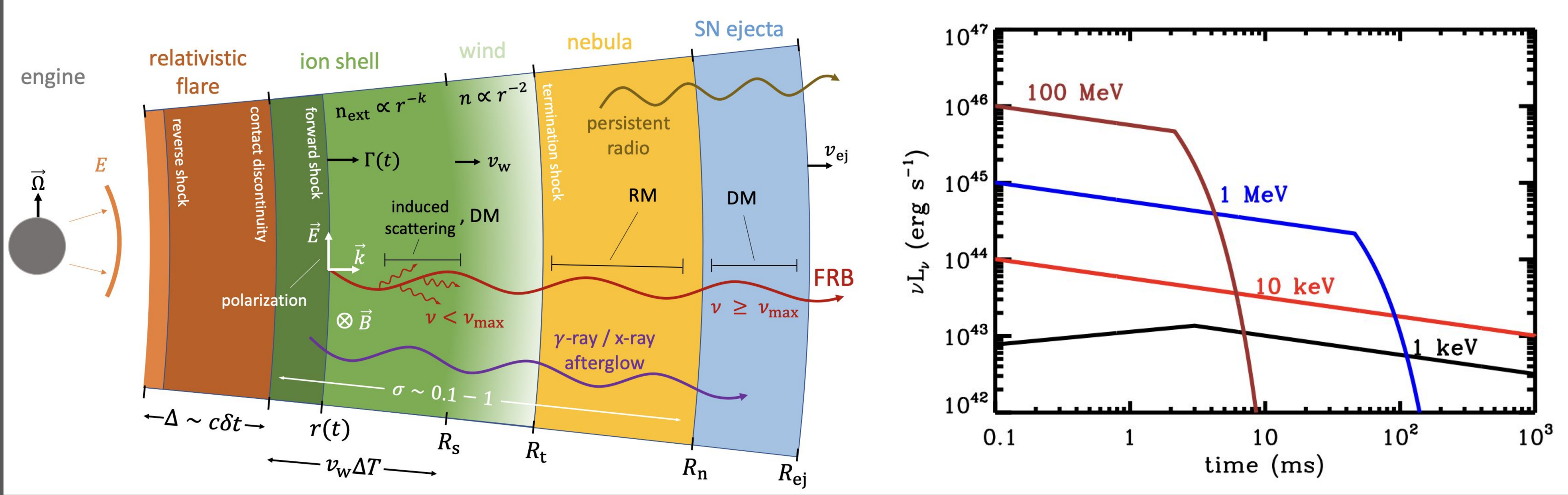
Abstract

The Very Energetic Radiation Imaging Telescope Array System (VERITAS) is typically used to observe in the Very-High-Energy (VHE) band of 100 GeV or greater. We demonstrate that by calculating per-pixel pedestal variance across the entire field of view, **VERITAS will be capable of detecting extended "wavefront" emission in the MeV-GeV range lasting 10s of microseconds or longer.** Some models of Fast Radio Burst (FRB) emission predict very weak emission in this energy range, and we aim to develop a trigger sensitive enough to search for this counterpart emission.

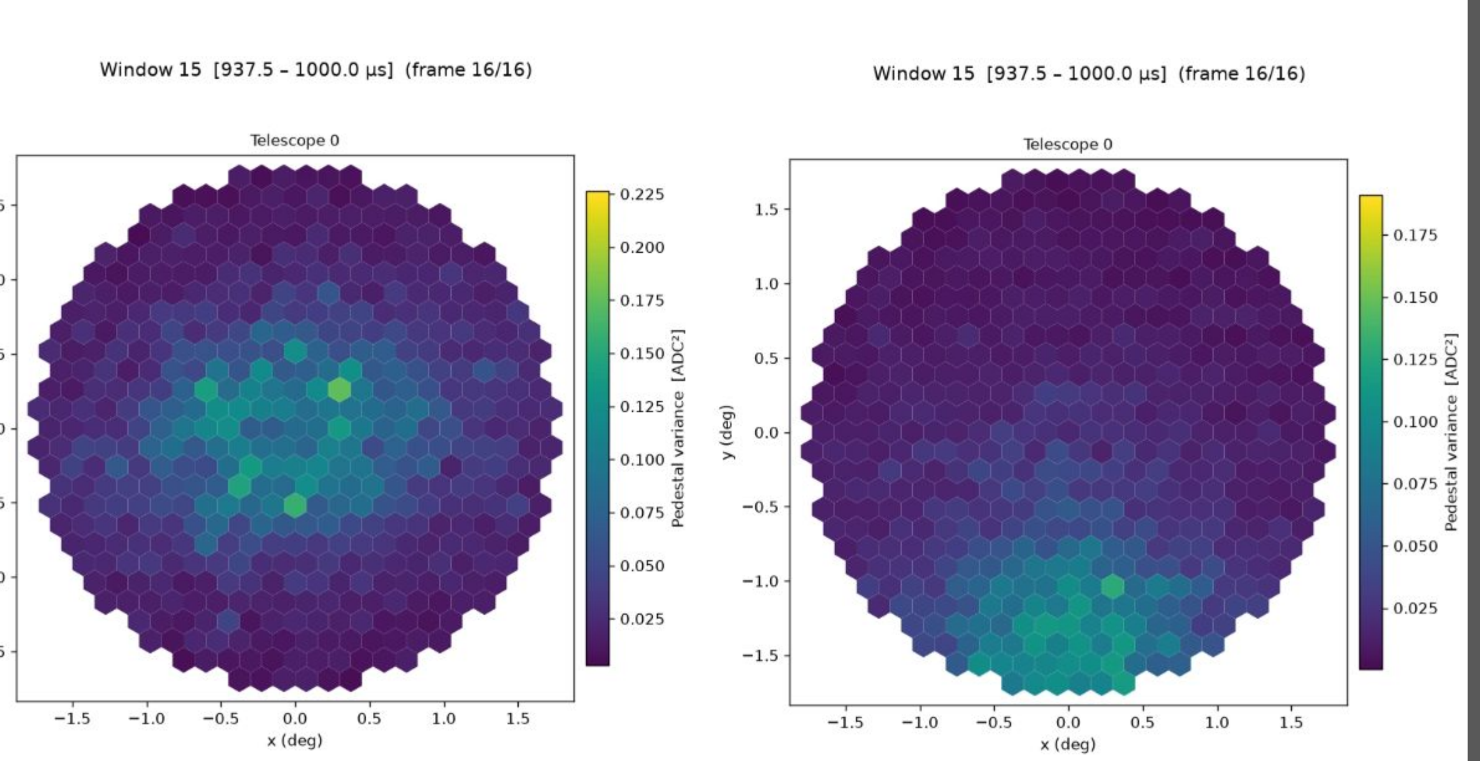
MeV-GeV Counterparts as Predicted by Metzger et. al (2019) PIC Simulation



Motivation: The Synchrotron Maser Afterglow

There are many candidate theories to explain the mechanism behind Fast Radio Bursts, but there are some that explicitly predict counterpart emission in energies greater than 10s of MeV. In particular, if FRBs are explained as ultra-relativistic magnetized shocks (which is known as a synchrotron-maser model), thermal electrons would be heated by the shock, causing them to briefly glow in the X-ray and γ-ray bands. This "afterglow" is expected to last 0.1-10ms and have a fluence of photons below 10^-2 photons/m^2. **Seeing such a short, low-fluence burst presents challenges that VERTIAS is uniquely able to address** with the Optical Monitoring Upgrade due to its 500Mhz raw sample rate, very large collection area, and field-of-view overlap with the CHIME FRB monitor.

1ms, 1.2 γ-ray/m^2 wavefront with no NSB

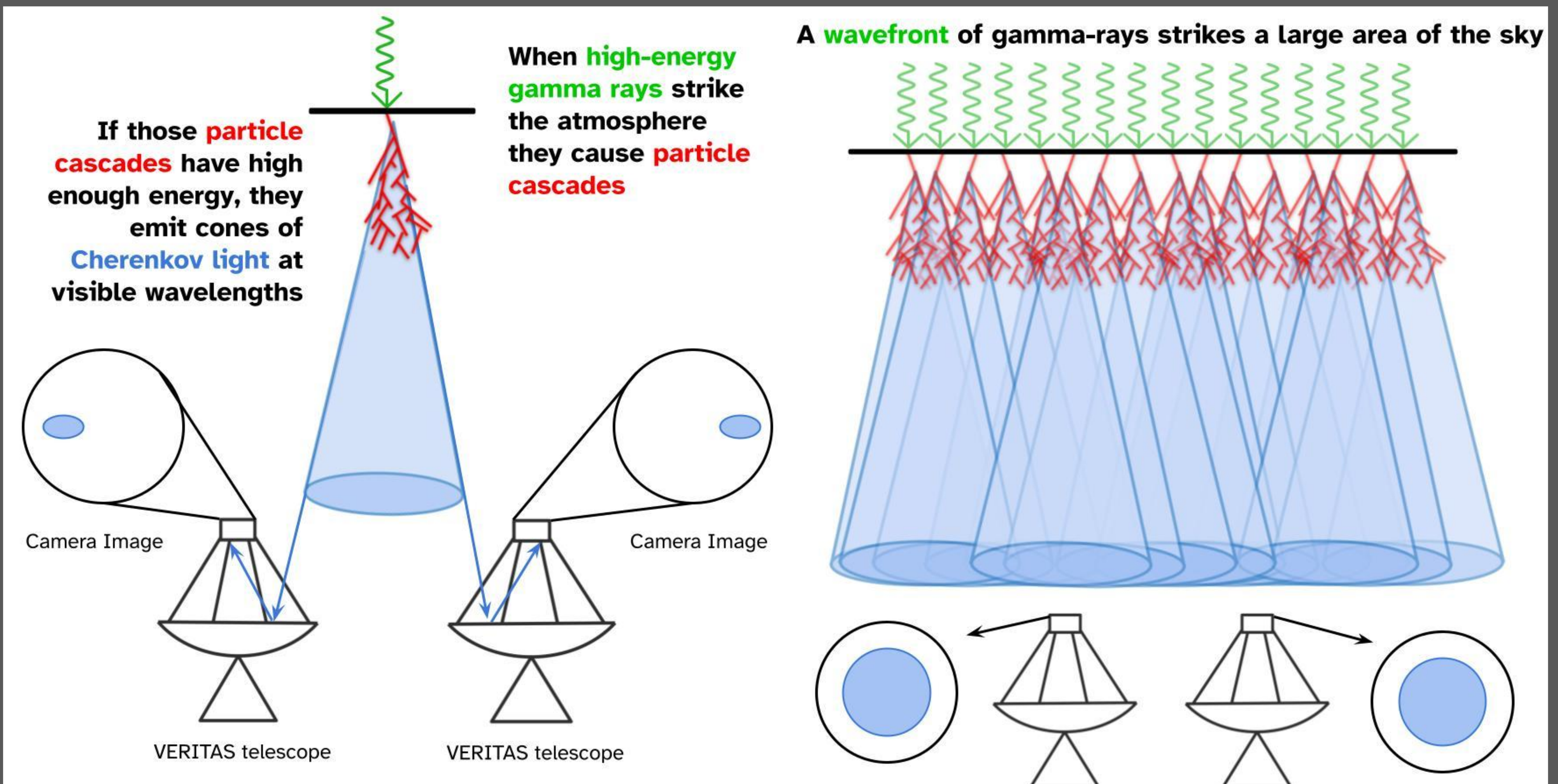


0 degree offset **1.5 degree offset**

GitHub

See these plots come alive!

The VERITAS array at Fred Lawrence Whipple Observatory in Arizona



Single rays create parallactic, small images **Wavefronts create identical, diffuse images**

Typical VHE Observation Versus Wavefront Observation

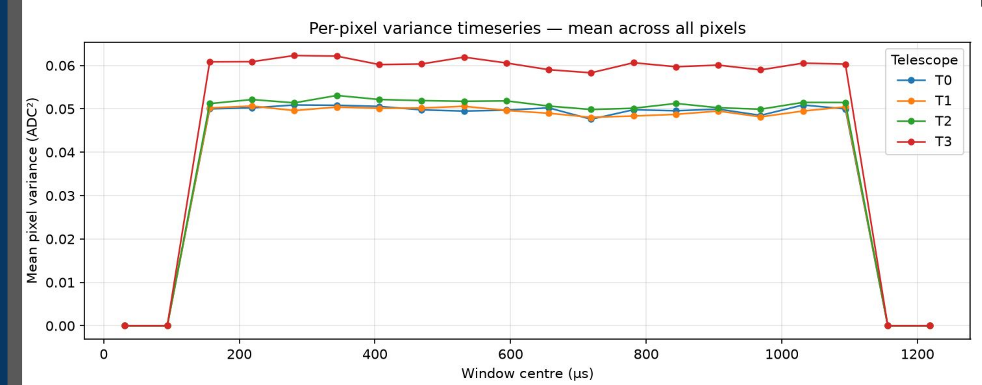
New Software

We develop a modified simulation pipeline, **Veritas-Wavefront**, that adds the ability to simulate the arrival of many primary gamma-ray photons over a wide area within a short time. CORSIKA and GrOptics simulate individual events as before, but now **across a configurable radius** (typically 1km). A new Python script uses **uproot** to read the GrOptics output files and creates a merged file with all the Cherenkov photons that reach a camera, but randomly scatters those arrival times **across a configurable length of time** to reflect the wavefront duration (typically 0.1-10ms). This file is passed to CARE to simulate the camera response. A typical VERITAS analysis relies on triggers to record select FADC samples. In our pipeline, we instead split the CARE output into **windows of configurable duration** and **compute the pedestal variance in each** [simply mean(samples^2) - mean(samples)^2]. This pedestal variance calculation simulates the FPGA-based Optical Monitoring Upgrade currently under development. From there, data can be visualized with basic tools like matplotlib rather than the relative complexity of VERITAS's EventDisplay.

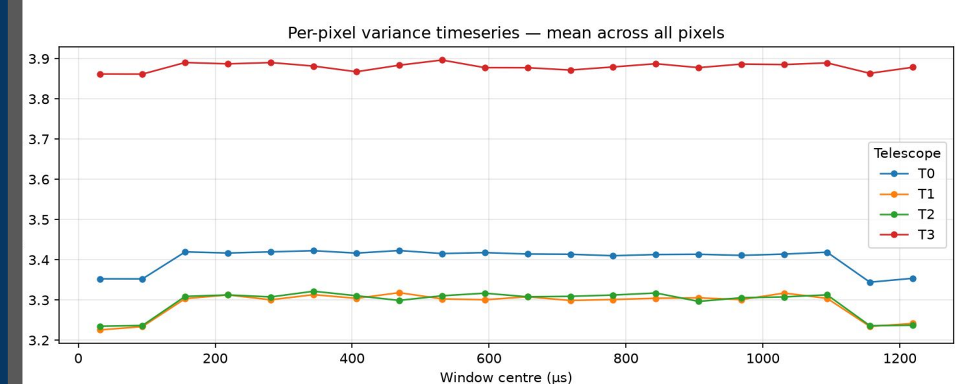
Existing Software

- **CORSIKA** simulates primary γ-rays and their particle cascades
- **GrOptics** traces cherenkov light from their emission through the mirrors and lenses
- **CARE** simulates the camera response and electronics.
- These are all used in the **VTS-SimPipe** GitHub repository for simulation.

Avg. pedvar for a strong wavefront:



No Night Sky Background (NSB)



Against NSB: slight, but detectable!

Next Steps

- Diagnose SNR issue in T4 — is this artificial or real?
- Establish sensitivity curves and dependence on zenith angle, offset angle, time of year, etc.
- Develop a detection metric better than avg. camera variance
- Incorporate the FoV overlap with CHIME into detection statistics

References:

Metzger, B. D., Margalit, B., & Sironi, L. 2019, "Fast radio bursts as synchrotron maser emission from decelerating relativistic blast waves," MNRAS, 485, 4091. doi:10.1093/mnras/stz700

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